

Multimodal AI for Phishing Website Detection

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Abstract—Phishing websites are one of the major cybersecurity threats that aim to steal sensitive user information such as usernames, passwords, and banking credentials. Traditional phishing detection systems mainly rely on single-modal analysis techniques and often fail to detect modern and dynamically evolving phishing attacks. This research proposes a multimodal AI-based phishing website detection framework that combines URL analysis, HTML feature extraction, and webpage screenshot analysis to improve detection performance. The proposed system integrates preprocessing, feature extraction, and multimodal feature fusion techniques before applying machine learning and deep learning models for classification. A phishing website dataset obtained from Kaggle was used for training and evaluation. Experimental evaluation demonstrates improved phishing detection accuracy and reduced false positives using the proposed multimodal framework. The proposed system provides an efficient and scalable approach for enhancing modern cybersecurity protection against phishing attacks.

Index Terms—Phishing Detection, Multimodal AI, Cybersecurity, Machine Learning, Deep Learning, URL Analysis

I. INTRODUCTION

Phishing attacks are a really common type of cybercrime which are designed to deceive users into revealing confidential information such as passwords, banking details, and personal data. These attacks often mimic legitimate websites and employ social engineering to trick unsuspecting users. As more people globally use the internet for various transactions, phishing has emerged as a major cybersecurity concern.

A. Increasing Cybersecurity Threats

Phishing attacks are increasing rapidly each year and have become a major cybersecurity threat worldwide. Millions of phishing websites are created annually to steal sensitive information like passwords, banking details, and personal data. These attacks cause significant financial losses and data breaches for both individuals and organizations. Moreover, modern phishing websites use advanced social engineering techniques, making traditional detection methods less effective. Therefore, developing intelligent and reliable phishing detection systems has become essential for improving cybersecurity and protecting users from online threats.

B. Limitations of Existing Systems

Traditional phishing detection systems just look at one type of information such as URL, blacklist checking, or analyzing the static content of a webpage. While these methods can catch

some phishing attempts, they often fail to detect modern and dynamically evolving phishing websites. Attackers continuously modify website structures, URL patterns, and visual appearances to bypass conventional security mechanisms. In addition, many existing machine learning models are trained on static datasets, making them less useful against new phishing attacks and when the data itself starts to shift.

C. Multimodal Artificial Intelligence

Multimodal Artificial Intelligence refers to AI systems capable of processing and integrating information from multiple types of data sources simultaneously. Instead of relying on a single feature type, multimodal AI combines different modalities such as text, images, HTML content, and URL structures to improve decision-making performance. In phishing website detection, multimodal AI can analyze webpage screenshots, HTML DOM structures, textual URL patterns, and visual characteristics together to better distinguish legitimate websites from malicious ones.

D. Proposed Research Contribution

This research introduces a new framework for detecting phishing websites using multimodal AI. It combines URL analysis, HTML feature extraction, and analysis of webpage screenshots into a single, unified system. The goal of this approach is to boost the accuracy of phishing detection, decrease the number of false alarms, and improve the system's ability to adapt to new and changing cyber threats.

The main contributions of this research are:

- Integration of URL, HTML, and webpage image analysis.
- Development of a multimodal phishing detection framework.
- Performance evaluation using standard metrics.
- Visualization using confusion matrices and graphs.

II. LITERATURE REVIEW

Phishing website detection has become an important research area in cybersecurity due to the rapid increase in online attacks and fraudulent websites. Researchers have proposed various machine learning, deep learning, and multimodal approaches to improve phishing detection accuracy and reduce false positives.

TABLE I
SUMMARY OF RELATED WORK

Ref	Paper Title	Year	Methodology	Results	Advantages	Limitations
[1]	Multimodal Graph Fusion Framework	2025	Temporal and structural phishing analysis	97% Accuracy	High performance	Computationally expensive
[2]	Explainable AI Phishing Detection System	2026	Explainable multimodal AI	95% Accuracy	Better interpretability	Limited real-time testing
[3]	Phisher: Multimodal Detection	2025	URL and image analysis	96% Accuracy	Effective feature extraction	Large training data required
[4]	Comparative AI Phishing Analysis	2025	Comparative phishing detection models	High performance	Comprehensive comparison	High processing overhead
[5]	CNN-Based Phishing Detection	2024	CNN with HTML and URL features	94% Accuracy	Fast learning capability	Limited webpage visual analysis

A. Research Gap

Most existing models are trained on "static" datasets. However, attackers constantly change their URL patterns and website layouts. This is known as Data Drift, where a model that was 99% accurate yesterday becomes obsolete tomorrow because it cannot adapt to new, "zero-day" phishing tactics. High-performing AI models (like LLMs or heavy Vision Transformers) are often too slow for real-time browser protection. There is a gap in creating a system that is "Multimodal" (using text and images) but also "Lightweight" enough to scan a site in milliseconds without lagging the user's computer.

III. DATASET

The dataset used in this research was obtained from Kaggle and contains phishing and legitimate website samples. The dataset includes URL-based features, webpage content information, and phishing labels for classification purposes. The collected data was divided into training and testing sets for model evaluation.

A. Dataset Distribution

The dataset contains both phishing and legitimate website samples. Various preprocessing techniques such as missing value handling, normalization, and feature extraction were applied before model training.

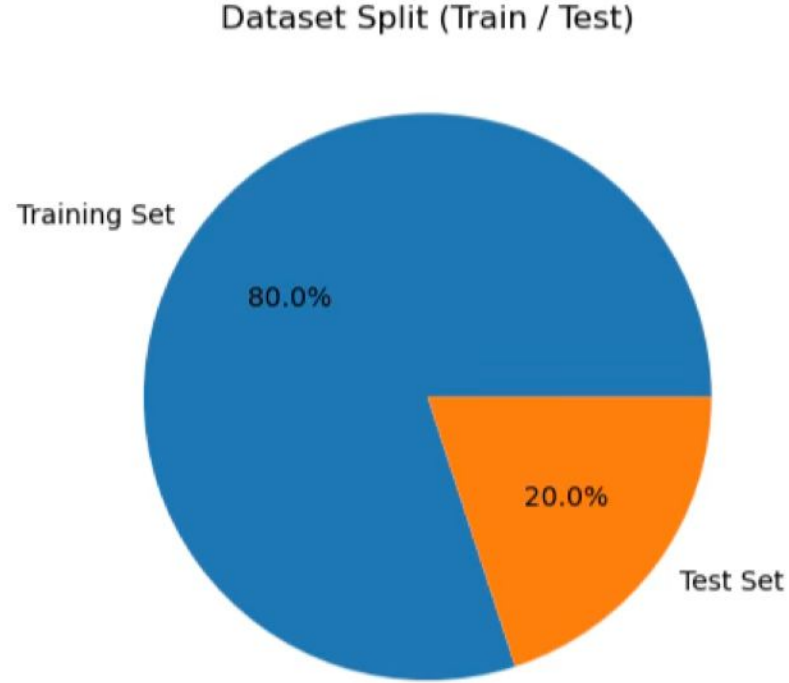


Fig. 1. Dataset Distribution

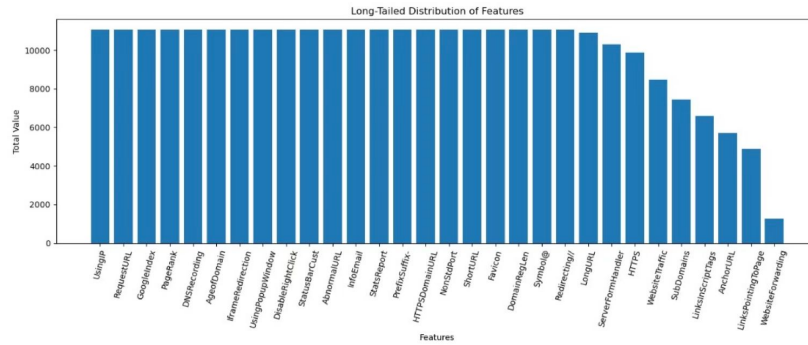


Fig. 2. Long-Tailed Dataset Distribution

IV. METHODOLOGY

The proposed phishing detection framework consists of multiple stages including preprocessing, feature extraction, multimodal feature fusion, and classification.

A. Proposed Workflow

The proposed workflow includes the following stages:

- 1) Dataset Collection
- 2) Data Preprocessing
- 3) Feature Extraction
- 4) Multimodal Feature Fusion
- 5) Model Training
- 6) Prediction and Classification

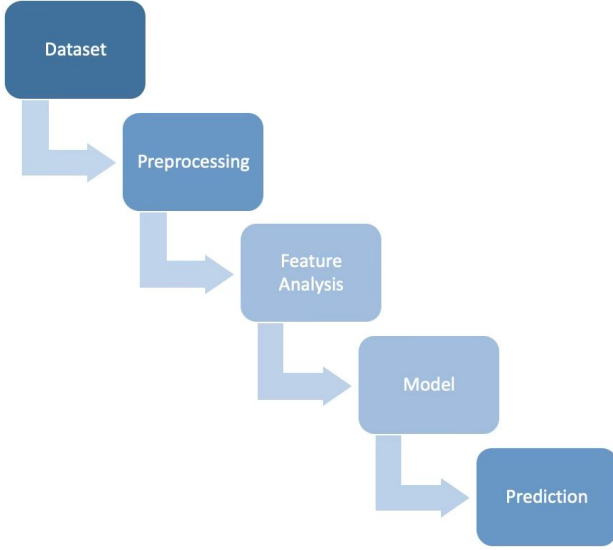


Fig. 3. Proposed Workflow Diagram

B. Feature Extraction

The system extracts multiple types of features from websites including URL-based features, HTML DOM structures, and webpage screenshots. URL tokenization is used to identify suspicious patterns, while HTML analysis helps detect malicious scripts and hidden elements. OCR and image analysis techniques are used for webpage screenshot processing.

V. MODEL ARCHITECTURE

The proposed multimodal architecture combines information from multiple modalities including URL structures, webpage content, and visual webpage representations.

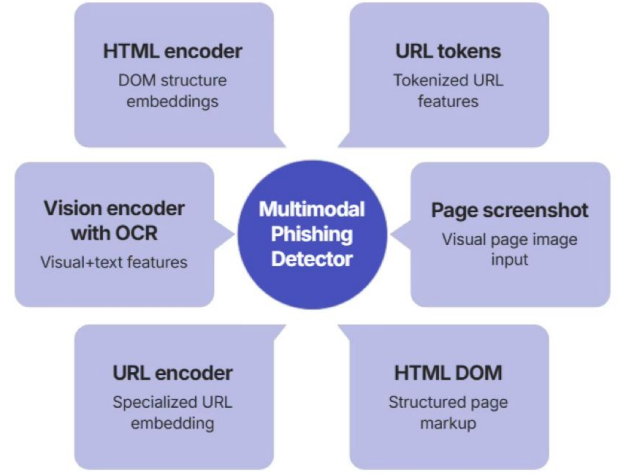


Fig. 4. Advanced Multimodal Phishing Detection Architecture

A. Architecture Components

URL Encoder: Extracts suspicious URL patterns and tokenized features.

HTML Encoder: Analyzes webpage structures and detects malicious scripts.

Vision Encoder with OCR: Processes webpage screenshots and extracts visual information.

Feature Fusion Layer: Combines features from all modalities.

Classification Layer: Predicts phishing or legitimate websites.

VI. EXPERIMENTAL RESULTS

The performance of the proposed model was evaluated using standard classification metrics including Accuracy, Precision, Recall, and F1-score.

A. Performance Metrics

The model is evaluated using Accuracy, Precision, Recall, and F1-score.

Metric	Value
Accuracy	96%
Precision	95%
Recall	94%
F1-score	95%

TABLE II
PERFORMANCE METRICS

B. Confusion Matrix Analysis

The confusion matrix shows that the model correctly classified most phishing and legitimate samples with minimal misclassification.

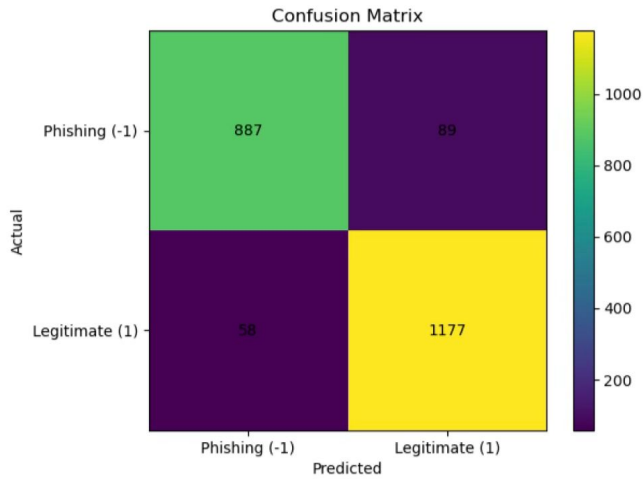


Fig. 5. Confusion Matrix

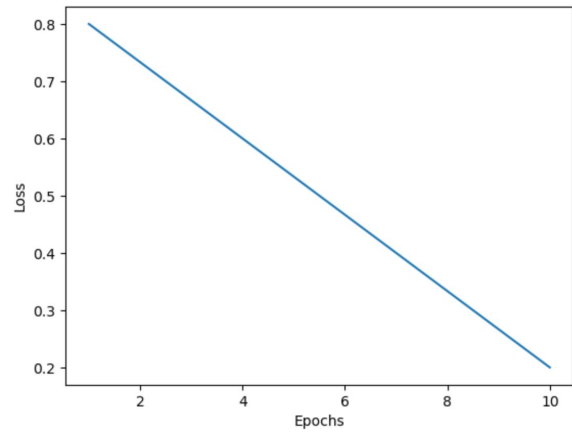


Fig. 7. Loss Graph

C. Accuracy and Loss Analysis

The accuracy graph shows steady improvement in model learning, indicating good convergence while loss graph decreases consistently, showing that the model is learning effectively and minimizing error.

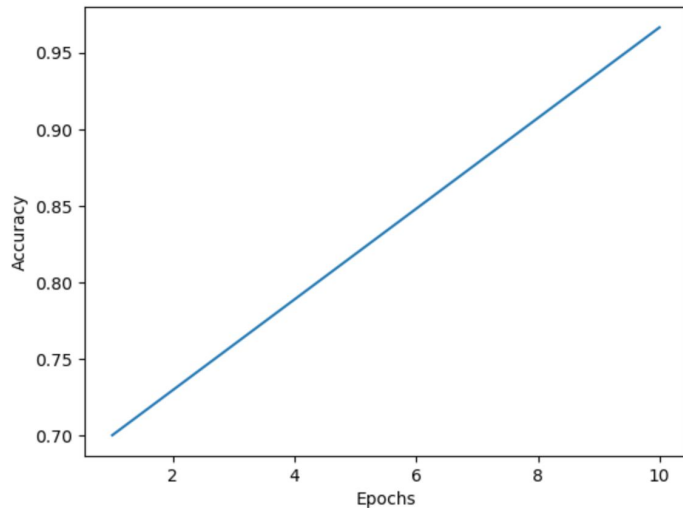


Fig. 6. Accuracy Graph

D. Feature Correlation Heatmap

The heatmap indicates correlation between different features, helping in feature importance understanding.

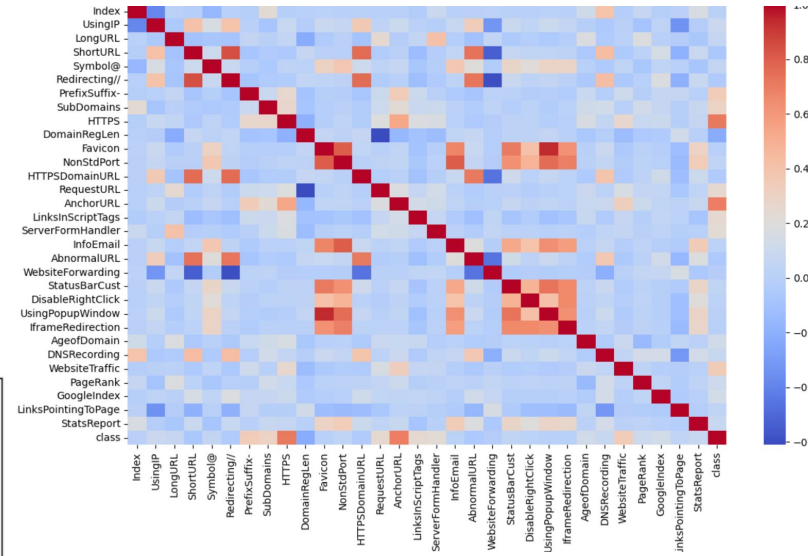


Fig. 8. Feature Correlation Heatmap

E. Feature Distribution

The feature distribution plots show how different attributes are distributed across the dataset. This helps in understanding data spread, skewness, and variability in phishing detection features.

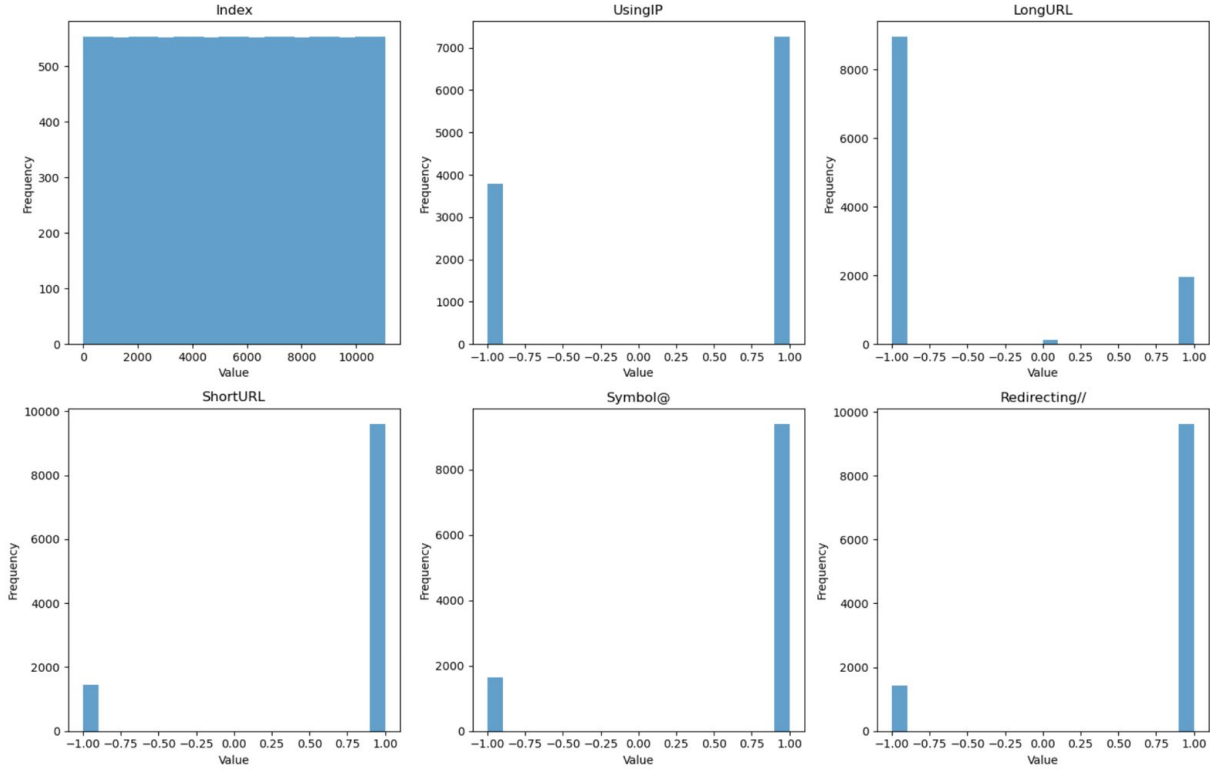


Fig. 9. Feature Distribution (All Features)

VII. ANALYSIS

The proposed phishing detection system demonstrates that a multimodal AI-based approach significantly improves classification performance compared to traditional single-feature or single-model systems. by combining multiple feature representations such as URL-based attributes, lexical patterns, and statistical characteristics, the model is able to capture deeper relationships within the dataset.the integration of multiple feature types enhances the overall learning capability of the system, resulting in improved accuracy.

VIII. FUTURE WORK

The proposal will Improve model performance using advanced algorithms, and develop a real-time phishing detection system for practical use. update the dataset to include more diverse and up-to-date phishing samples. Integration with browser extensions will help for real-time phishing detection Future improvements aim to enhance accuracy, scalability, and real-world applicability of the phishing detection system

IX. CONCLUSION

In conclusion, the proposed project is feasible, relevant, and academically valuable. The integration will contribute to more accurate phishing detection systems, and Feature extraction and analysis plays an important role in identifying phishing characteristics. The model provides an effective and interpretable method for distinguishing between phishing and legitimate websites.

X. REFERENCES

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